(iii) $-1, 0, 2, -2, 3, 6, -3, 5, 1, 4$

(iv) $-9, 6, 5, -3, 4, 2, -7, 6, -5, 10, -11, 0, 1$

## Spanning Tree :-

Let $G_1$ be a connected graph. A subgraph $T$ of $G_1$ is called a spanning tree of $G_1$ if (i) $T$ is a tree and (ii) $T$ contains all vertices of $G_1$.

## Prefix codes and weighted trees

* Sequence is a set whose elements are listed in order as the first element, second element and so on.
* The number of elements contained in a sequence is called its length.
* A sequence consisting of only 0 and 1 is called binary sequence or a binary string.

01, 001, 101, 11001, 1000100
with lengths 2, 3, 3, 5, 7

* Binary sequences are used as codes for letters for message sent through transmitting channels.

Suppose message consisting of the letters a, e, n, r, t is to be transmitted.

a: 1, e: 0, n: 10, r = 01, t = 101.

Suppose a message eat is to be transmitted. Then 0110 1.
Can this binary sequence be decoded correctly?
Because this sequence can also be written as ent, rar or eaar.

Prefix codes:-

Let P be a set of binary sequences that represent a set of symbols. Then P is called a Prefix code if no sequence in P is the Prefix of any other sequence in P.

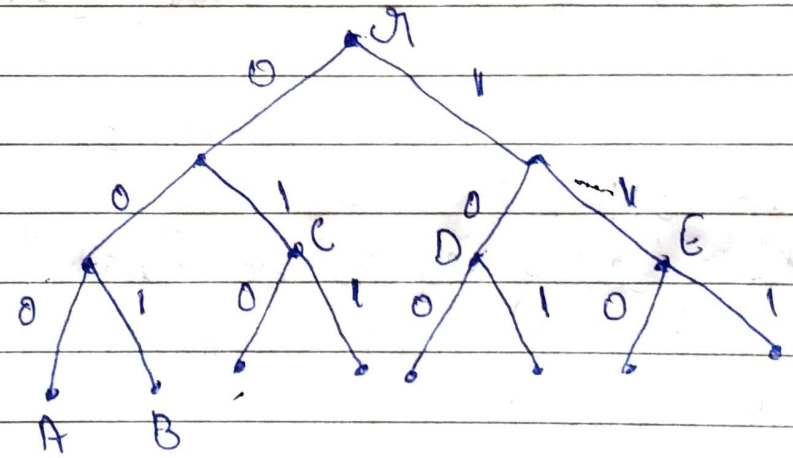
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$P_1 = \{10, 0, 1101, 111, 1100\}$
$P_2 = \{000, 001, 01, 10, 11\}$.

Prefix codes can be represented by binary trees

$$P_2 = \{000, 001, 01, 10, 11\}$$

* The longest sequence has length 3.
Construct a full binary tree of height 3




(1) Consider a the Prefix code
a: 111 b: 0 c: 1100 d: 1101 e: 10
using these code, decode the following sequence.
(i) 10011110 1 ii) 10 111 100 11000110
iii) 1101 111 1100 10

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Soln:-

(i) Splitting : 10 0 111 1101 Decode : e b a d

(ii) Splitting : 10 111 10 0 1100 0 1101
Decode : e a c b c b d.

(iii) Splitting 1101 111 1100 10 Decode : d a, c, c.

(2) A code for {a, b, c, d, e} is given by
a : 00 b : 01 c : 101 d : x10 e : y21.
where x, y, z ∈ {0, 1}. Determine x, y and z so that the given code is a Prefix code.

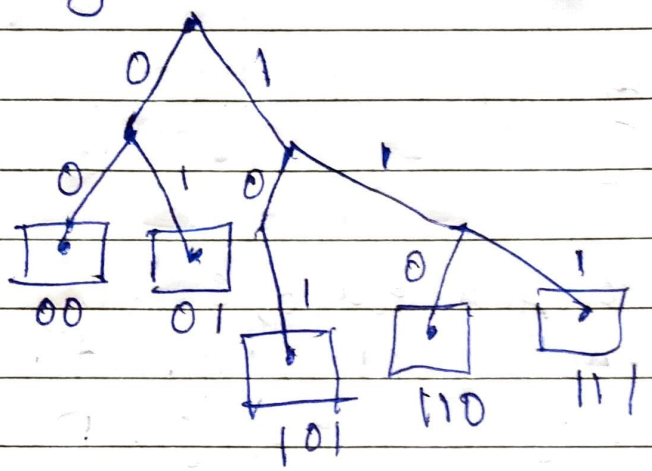
Soln:- x = 0 then the code for d = d0'
x = 1 d = 110.
Then we set y = 0, z = 0 · e = 001
which contains the code for a as a Prefix.
If y = 0, z = 1 e = 011, which contains the codes for b as a prefix.
If we set y = 1, z = 0 e = 101
y = 1, z = 1 e = 111
a = 00 b = 01 c : 101 d : 110 e : 111

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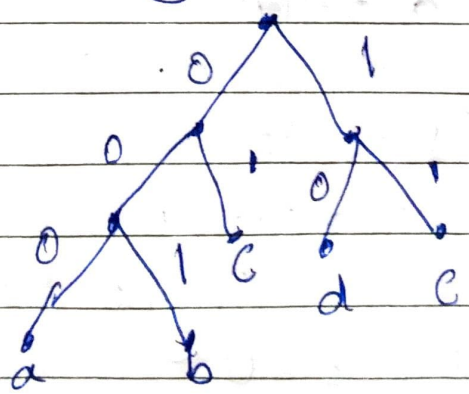
(3) Construct a binary tree that represents the Prefix code obtained

$$P = \{00, 01, 101, 110, 111\}$$

Soln:- length is 3.




(4) Obtain the Prefix code represented by the following labeled complete binary tree.



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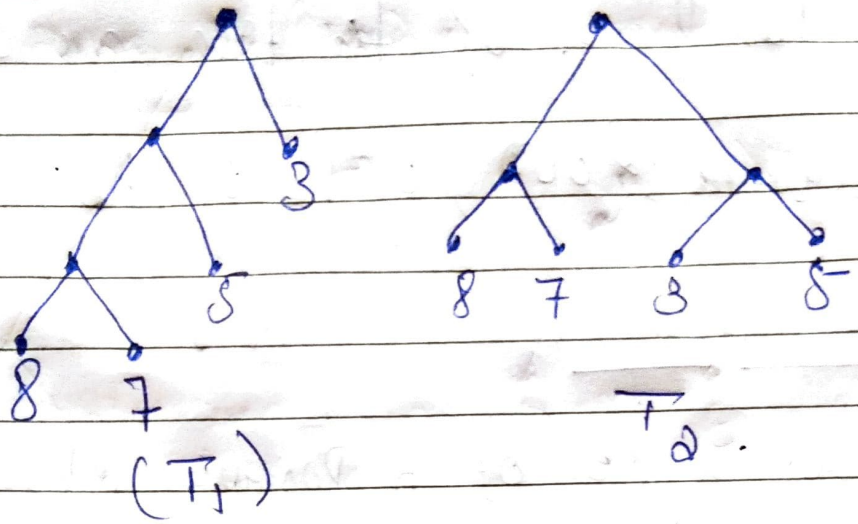
Soln :- The leaves of the tree are a, b, c, d, e

Sequence : {000, 001, 01, 10, 11}

## Weighted Trees :-

Consider a set of n positive integers $w_1, w_2 \ldots w_n$ where $w_1 \leq w_2 \leq \ldots \leq w_n$. Suppose we assign these integers to the n leaves of a complete binary tree $T = (V, E)$ in any one-to-one manner. The resulting tree is called a complete, weighted, binary tree with $w_1, w_2 \ldots w_n$ as weights. If $l(w_i)$ is the level number of the leaf of T to which the weight $w_i$ is assigned then $W(T)$ defined by

$$W(T) = \sum_{i=1}^{n} w_i l(w_i)$$ is called the weight of the tree T.




$(T_1)$ $T_2$.

$$W(T_1) = (8 \times 3) + (7 \times 3) + (5 \times 2) + (3 \times 1) = 58$$

$$W(T_2) = (8 \times 2) + (7 \times 2) + (3 \times 2) + (5 \times 2) = 46.$$

Optimal Tree :- Consider the set of all complete binary trees to whose leaves these weights. A tree in this set which carries the minimum weight is called an optimal tree.

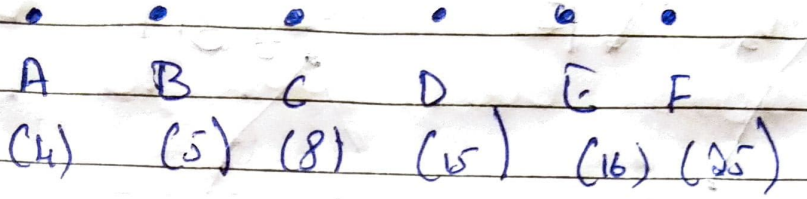
Construction of an optimal tree for a given set of weights is.

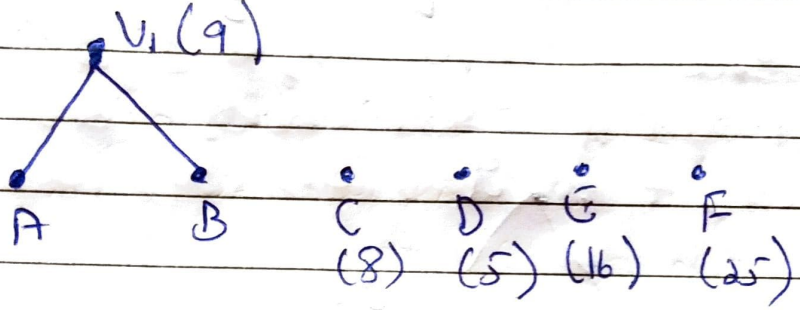
Let $\{4, 15, 25, 5, 8, 16\}$

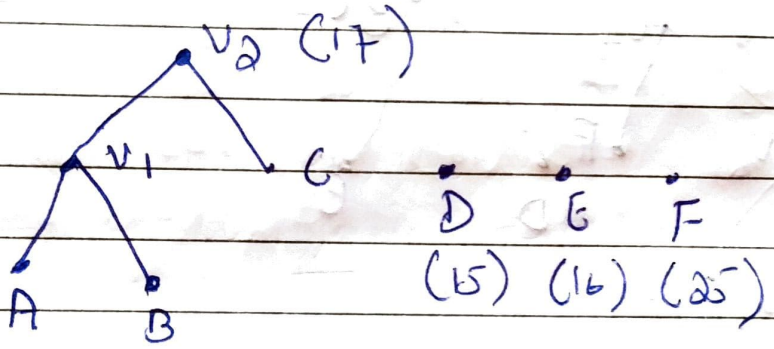
* arrange these six weights in a non-decreasing order and assign them to

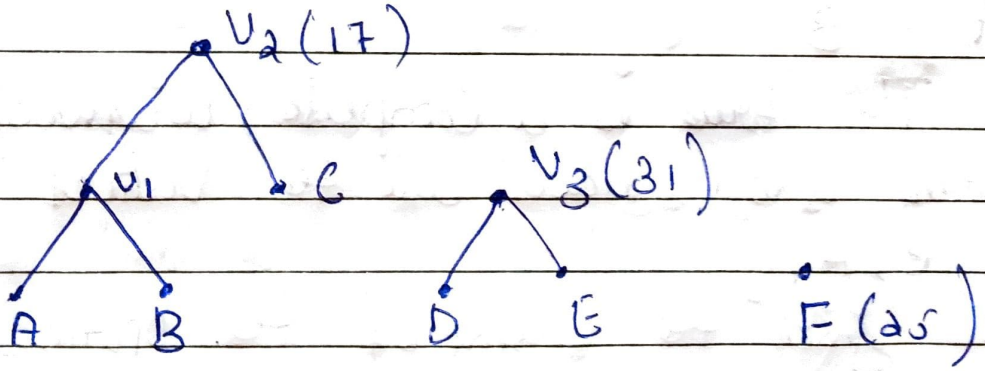
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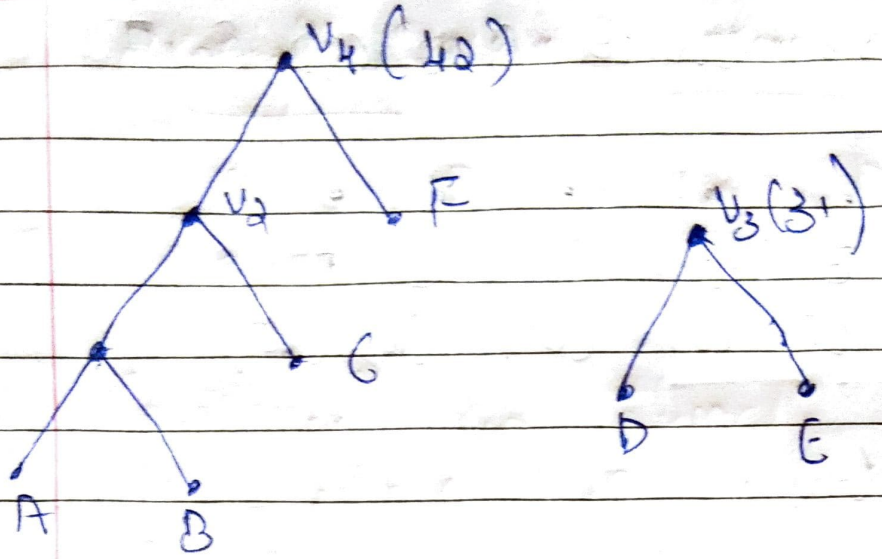
Six isolated vertices A, B, C, D, E, F

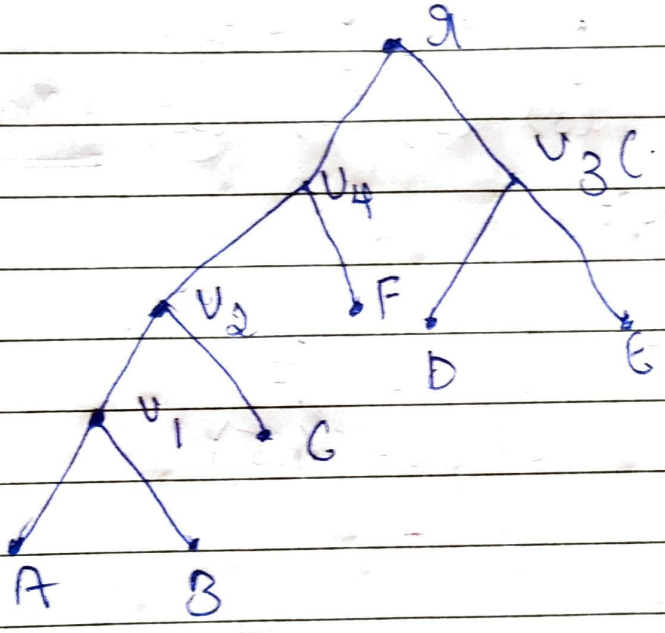





* The tree is a complete, weighted binary tree whose leaves are the vertices A, B, C, D, E, F.

To find the weight of the optimal tree.

$$W(T) = (4 \times 4) + (5 \times 4) + (8 \times 3) + (15 \times 2) + (16 \times 2) + (25 \times 2) = 172$$

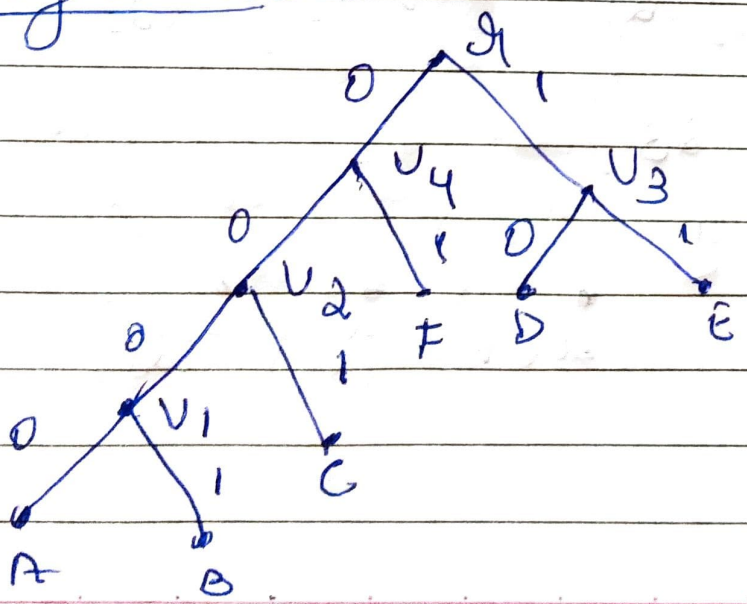
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The procedure that is used in constructing an optimal tree is known as Huffman's Procedure.

* The tree itself is often called an Huffman's tree.

## Optimal Prefix Code :-

* Huffman tree (optimal tree) can be used to obtain a prefix code for the symbols representing its leaves.
* The binary sequence through which the leaves are identified yield a prefix code for the symbols representing the leaves. This prefix code is known as an optimal prefix code.



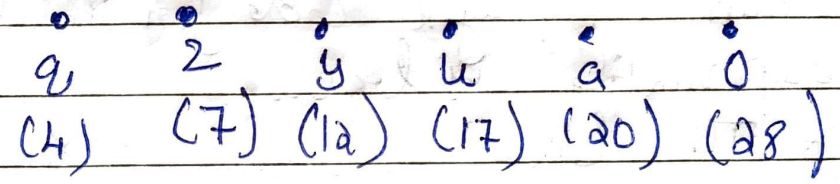
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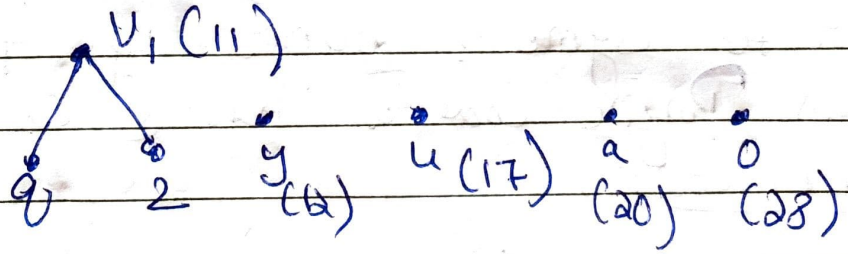
Binary Sequence of these code

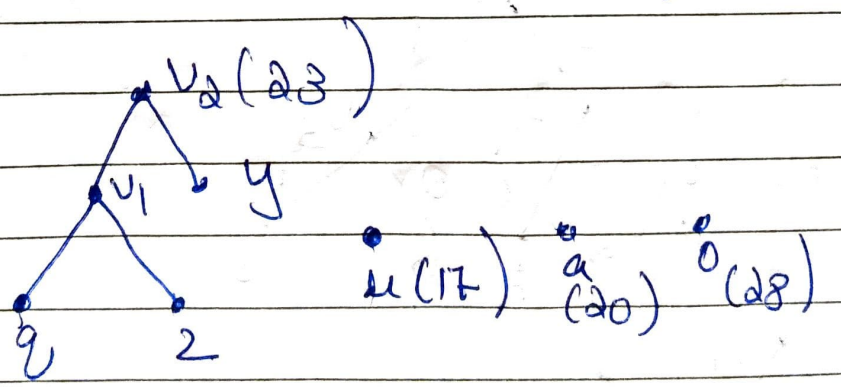
| A | B | C | D | E | F |
|---|---|---|---|---|---|
| 0000 | 0001 | 001 | 10 | 11 | 01 |

(2) Construct an optimal Prefix code for the Symbols a, o, q, u, y, 2 that occur with frequencies 20, 28, 4, 17, 12, 7.

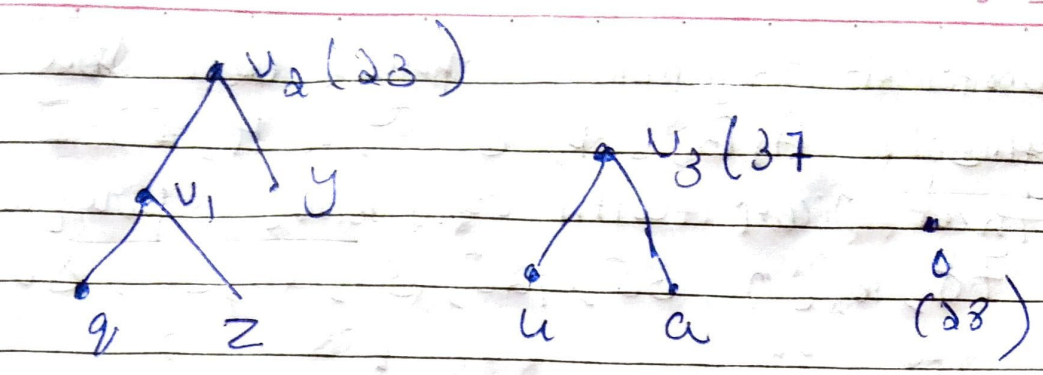
Soln:-

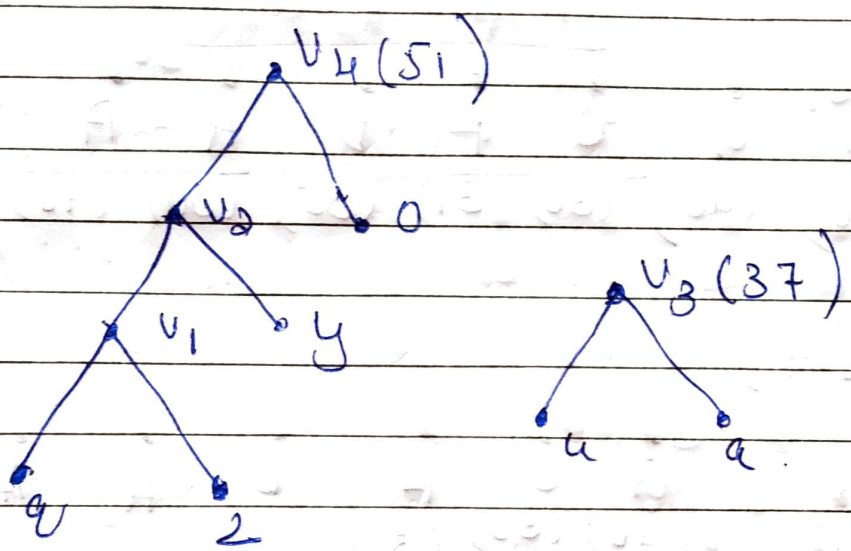


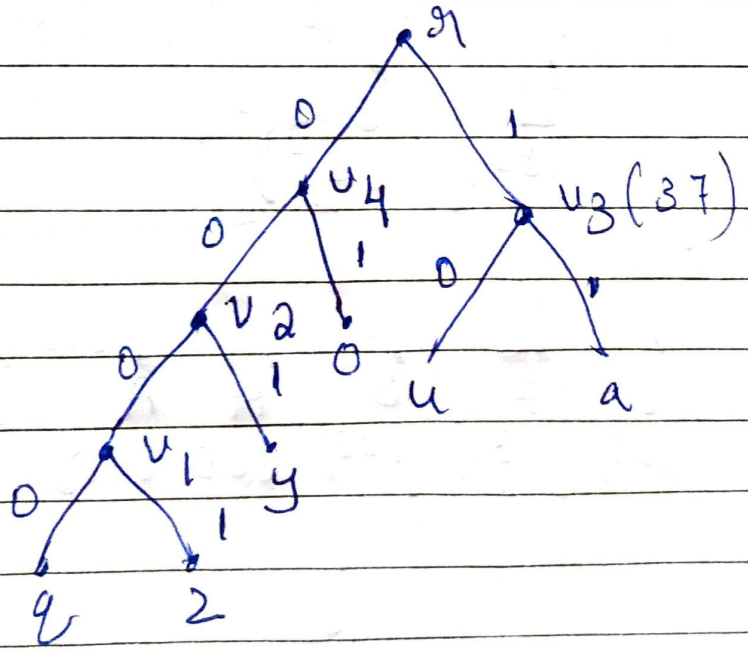



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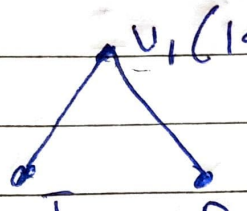




a : 11 0 : 01 q : 0000 u : 10
y : 001 2 : 0001.

Symbols A, B, C, D, E, F, G, H, I, J
that occur with respective frequencies
78, 16, 30, 35, 125, 31, 20, 50, 80, 3.

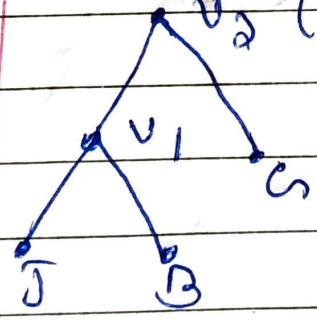
Soln :-

J B G C F D H A I E
(3) (16) (20) (30) (31) (35) (50) (78) (80) (125)



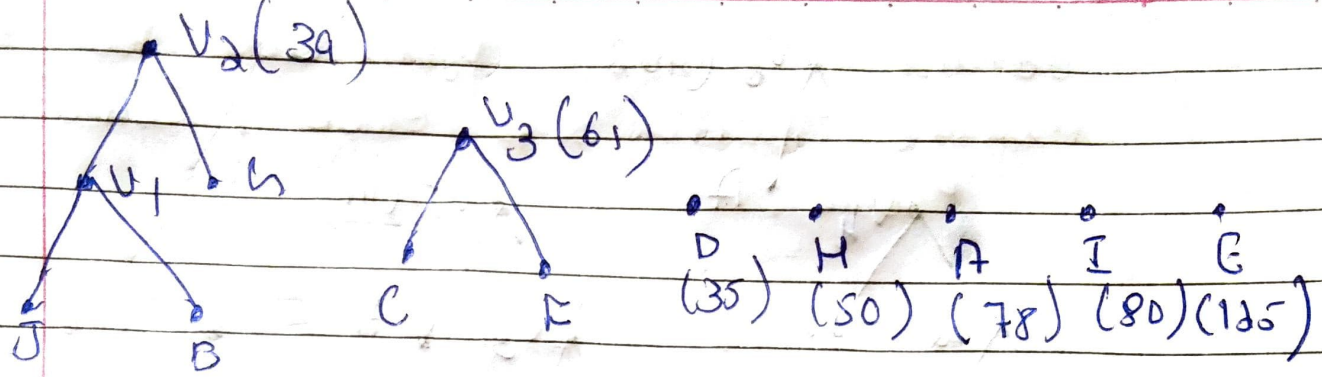

G C F D H A
(20) (30) (31) (35) (50) (78)

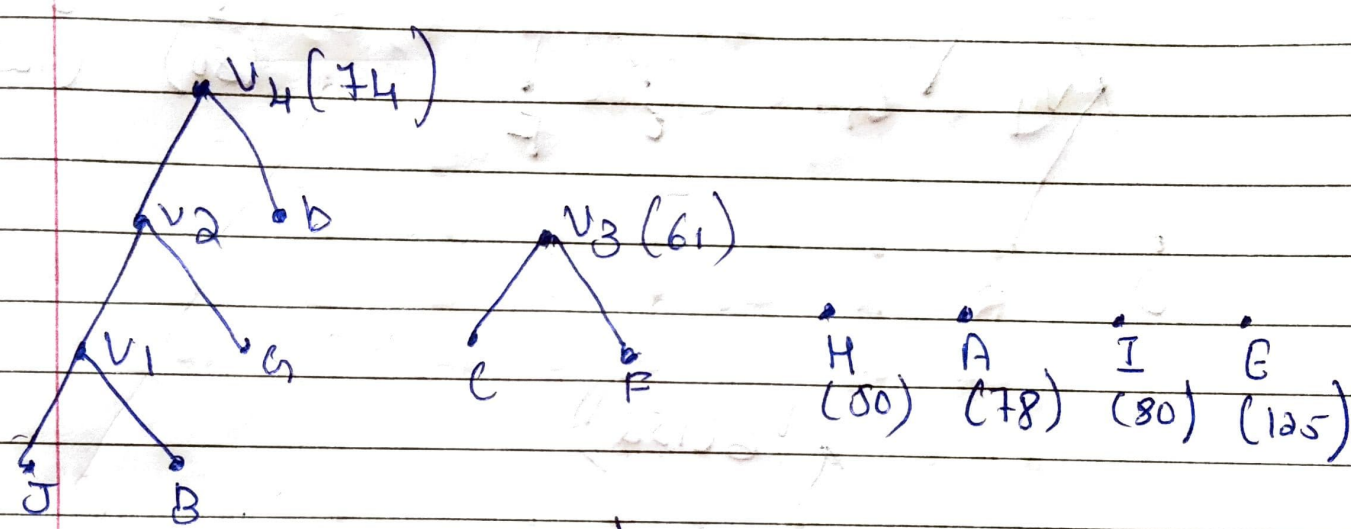
I E
(80) (125)

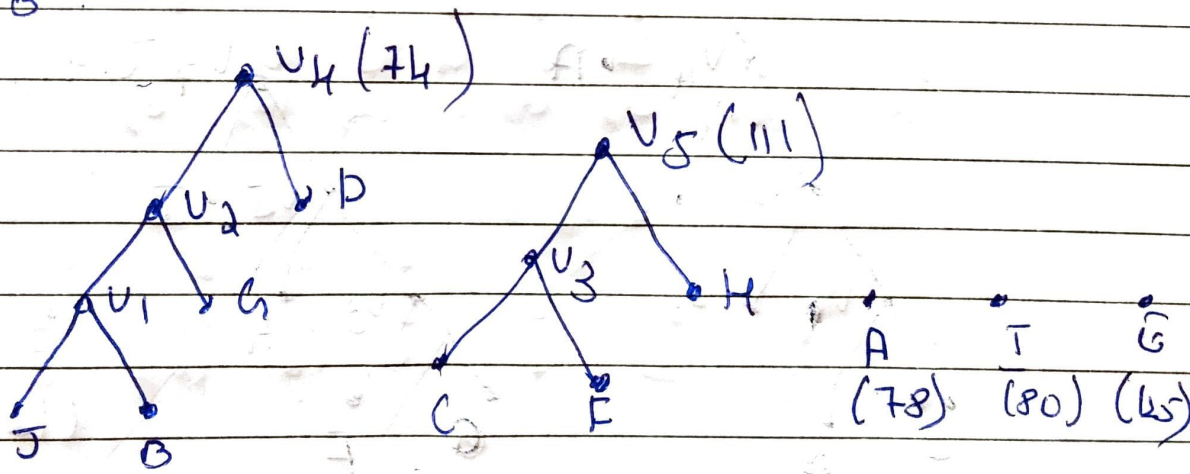


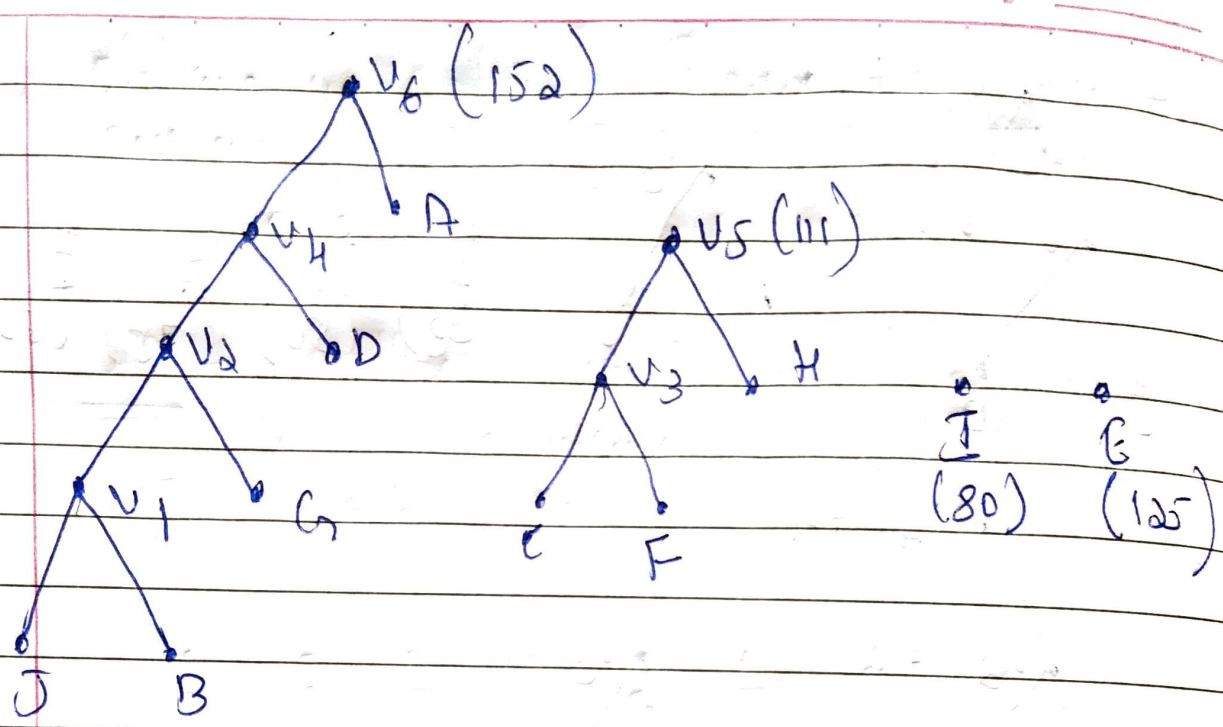

G F D H A I E
(30) (31) (35) (50) (78) (80) (125)

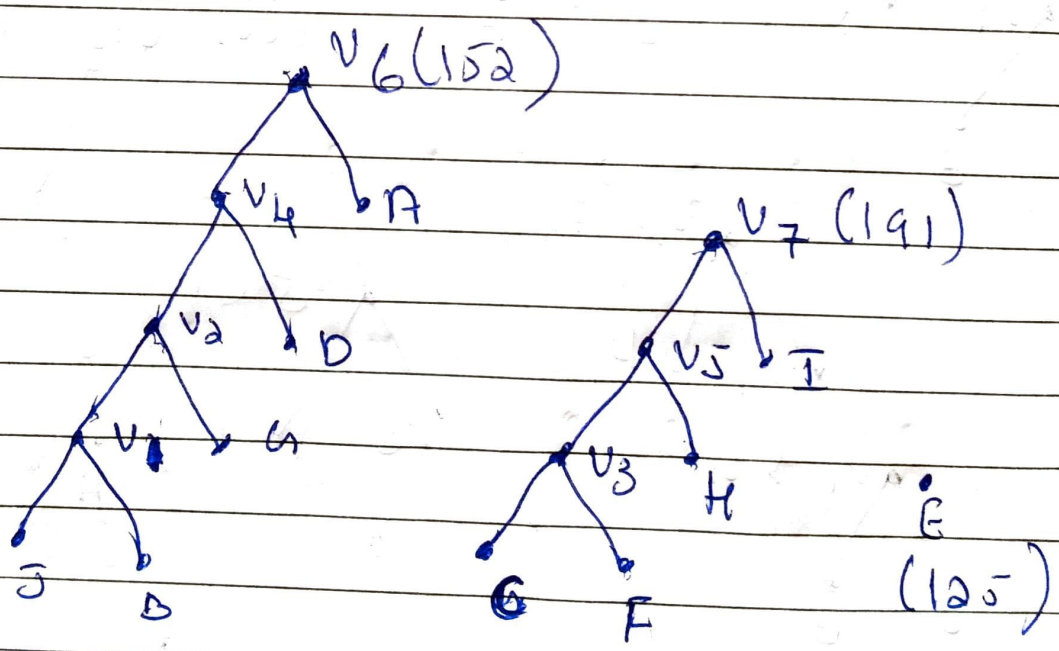
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V2(39)
V1
H
J
B
V3(61)
C
F
D (35)
H (50)
A (78)
I (80)
G (125)

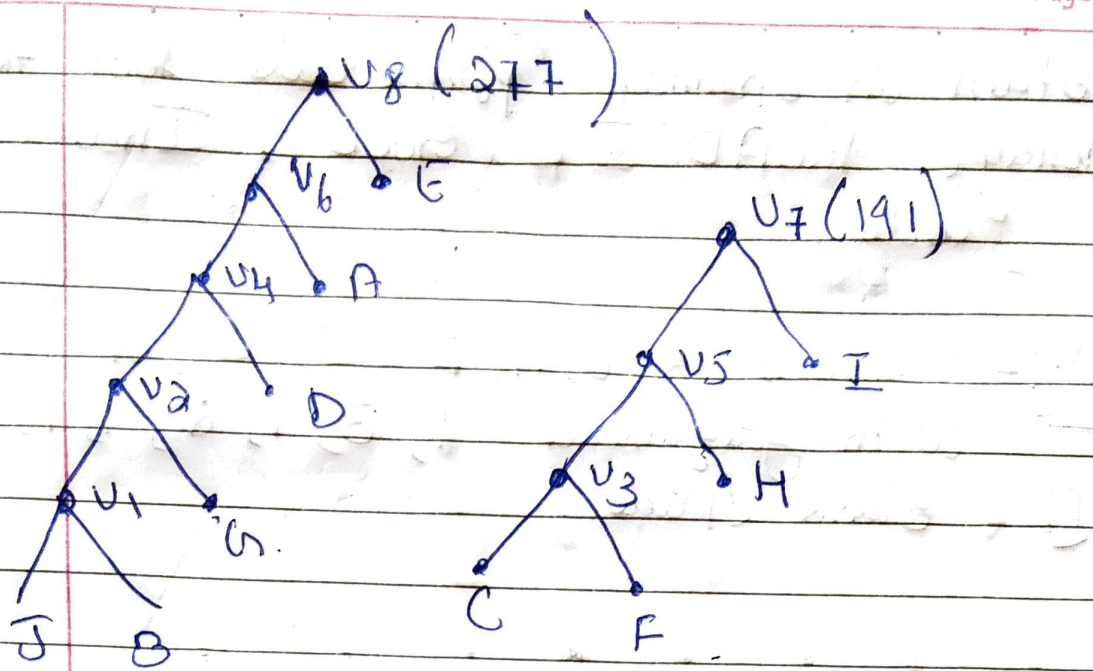

V4(74)
V2
D
V1
H
J
B
V3(61)
C
F
H (50)
A (78)
I (80)
G (125)

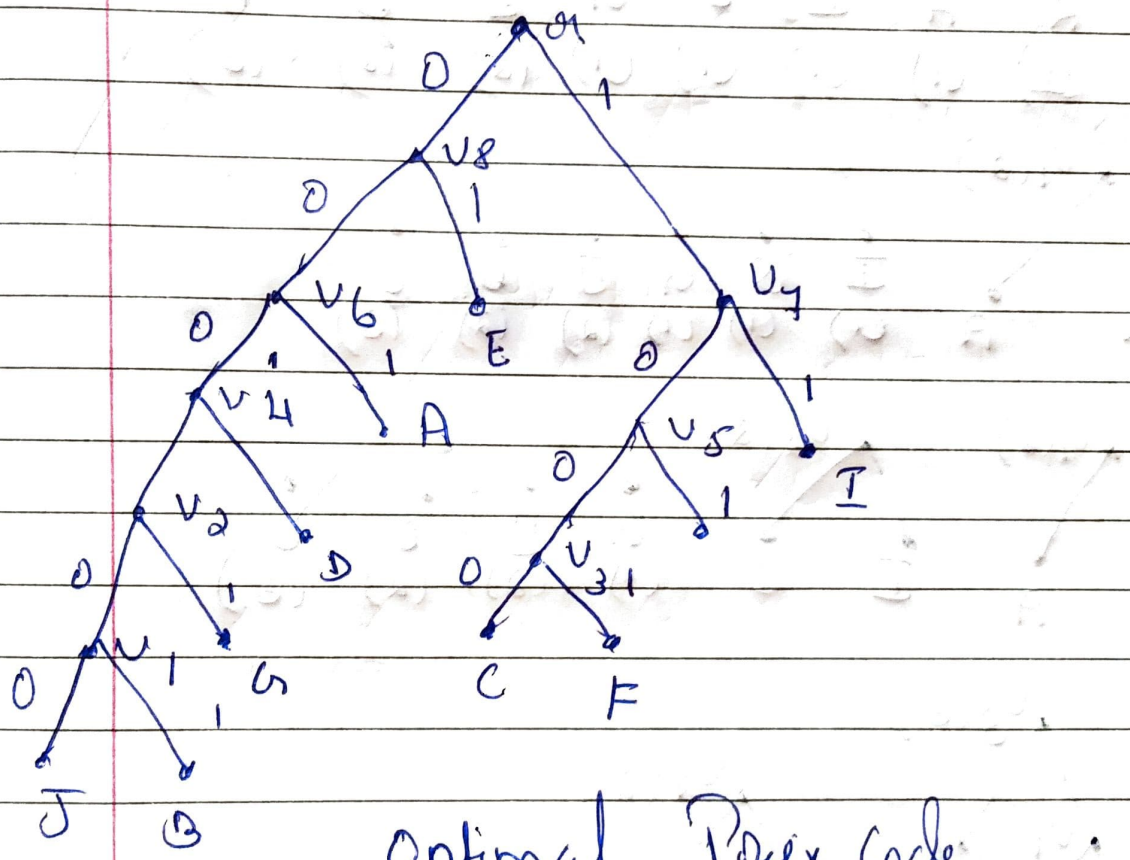

V4(74)
V2
D
V1
H
J
B
V5(111)
V3
H
C
F
A (78)
I (80)
G (45)


$v_6$ (152)
A
$v_4$
D
$v_2$
G
$v_1$
J
B
$v_5$ (111)
H
$v_3$
E
F
I
(80)
E
(125)


$v_6$(152)
A
$v_4$
D
$v_2$
G
$v_1$
J
B
$v_7$ (191)
I
$v_5$
H
$v_3$
E
F
G
(125)

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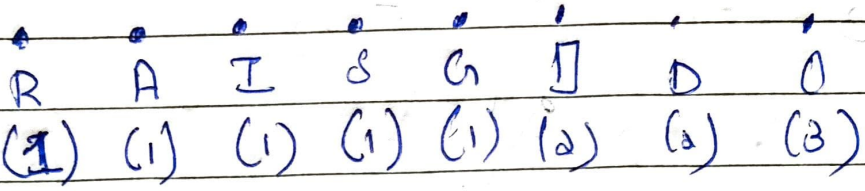

## Optimal Prefix Code.

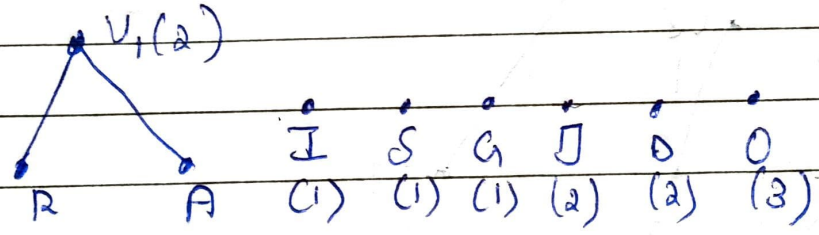
A : 001, B : 000001 C : 1000 D : 0001
E : 01 F : 1001, G : 00001, H : 101,
I : 11 J : 000000

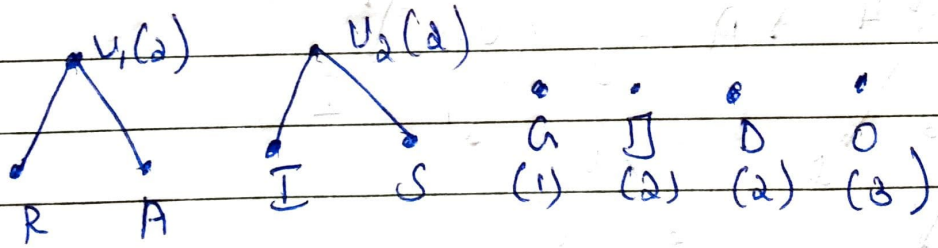
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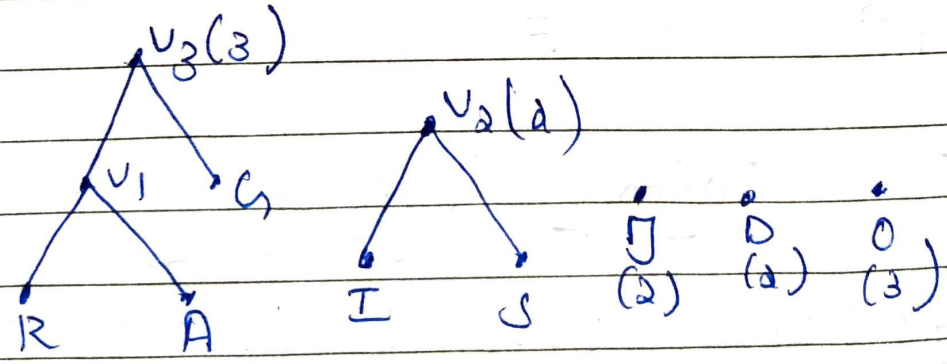
(4) Obtain an optimal Prefix code for the message ROAD IS GOOD. Indicate the code.

Soln:- Letters R, O, A, D, I, S, G, ␣ with frequencies 1, 3, 1, 2, 1, 1, 1, 1.
␣ - Blank space.

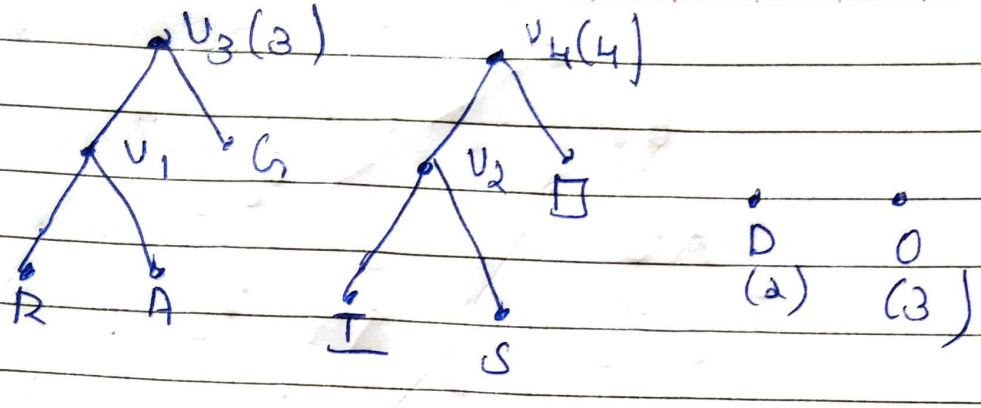


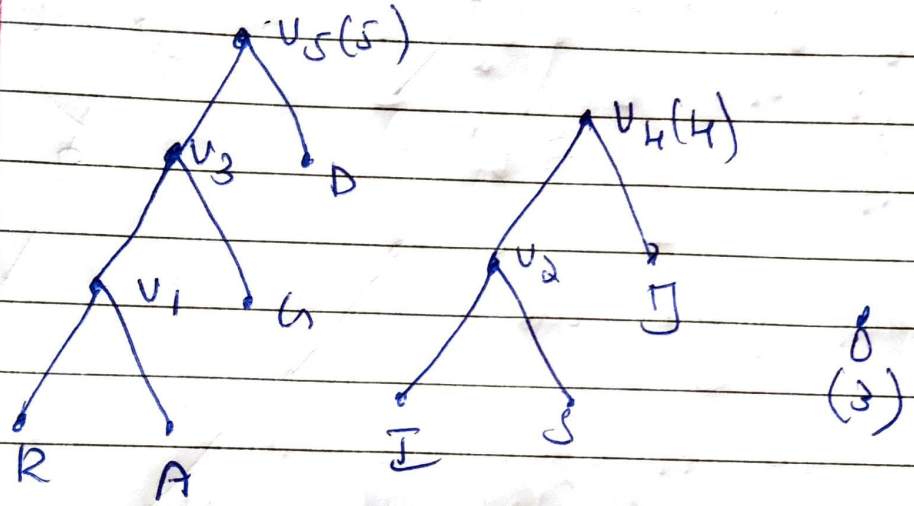



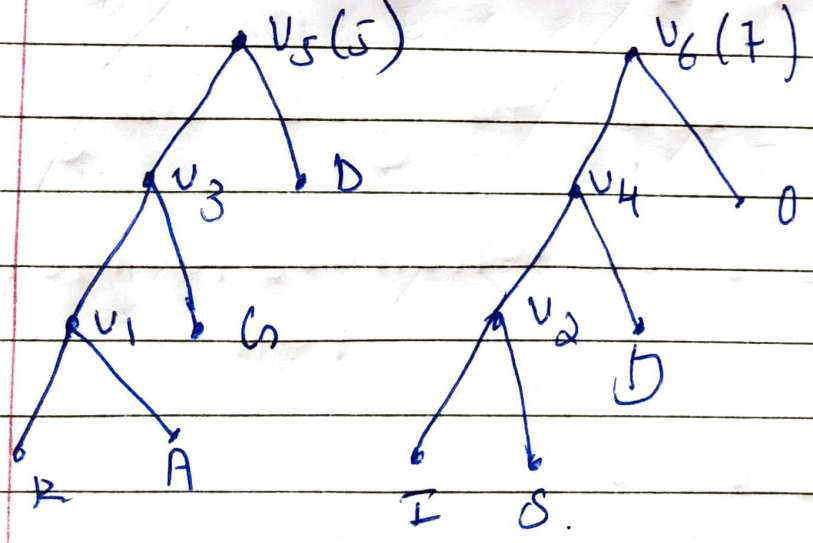



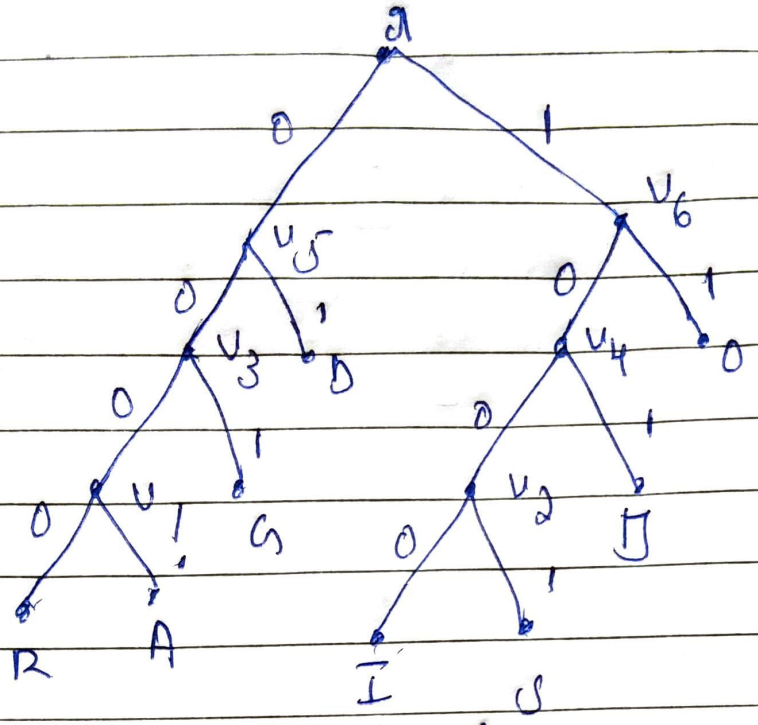



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$V_3(3)$
$V_1$
G
R
A
$V_4(4)$
$V_2$
I
I
S
D
(2)
O
(3)


$V_5(5)$
$V_3$
D
$V_1$
G
R
A
$V_4(4)$
$V_2$
I
I
S
O
(3)


$V_5(5)$
$V_3$
D
$V_1$
G
R
A
$V_6(7)$
$V_4$
O
$V_2$
I
I
S




Optimal prefix code

R : 0000 A : 0001 , I : 1000 , S : 1001
G : 001 O : 101 D : 01 U : 11

ROAD is GOOD for given message.

0000 11000 101 11 01 000 1001 101 00111
101.

(1)

| a | b | c | d | e | f | g |
|---|---|---|---|---|---|---|
| 10 | 30 | 5 | 15 | 20 | 15 | 5 |

(2)

| A | B | C | D | E | F | G | H |
|---|---|---|---|---|---|---|---|
| 22 | 5 | 11 | 19 | 2 | 11 | 25 | 5 |